

TASMail — Traceability & Integration Report

UI · API client · backend route · handler · feature integration
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1. Executive summary

TASMail is a webmail UI for any IMAP/SMTP server (BYOK). This report traces every user-facing surface of the production deployment at <https://mail.techatscale.io> from the React 19 SPA through the apiClient layer, into the Rust/Axum router, and down to the per-domain handler module. The matrix is produced from a static parse of router.rs (320 declared method+path pairs) and frontend/src/api/*.ts (56 client modules), then joined against component, hook, store, and util imports.

Backend endpoints (method + path)	320
Distinct feature groups	72
Frontend API client modules	56
Endpoints traced to a client	267 (83%)
Endpoints with no client (orphans)	53
Clients with no UI/hook/store consumer	2: sms-otp, snooze
Deployment target	https://mail.techatscale.io (live tunnel)
BYOK end-to-end (signup → IMAP → SMTP → folders)	PASS — proven by Playwright + raw HTTP

Headline findings:

- 83 % of declared backend endpoints are reachable from the SPA via an apiClient call. The remaining 17 % split into three groups: (a) genuinely backend-only — webhooks, SAML/OIDC callbacks, /metrics, /api/health, the public download token, the WebSocket handler, and the dedicated mobile API surface for the Flutter app; (b) admin features whose UI has not been built yet — cache controls, audit-log viewer, domain admin, payment-providers admin, user admin, IP warm-up admin; (c) niche flows (DANE verification GET, scheduled-message list, PST upload, push notification CRUD).
- Live BYOK chain proven twice: once through the UI via Playwright (Firefox, screenshots captured in frontend/e2e/screenshots/byok-noreply/), once via raw curl. Pre-BYOK /api/folders correctly 503s; post-IMAP-config it returns 5 folders and a non-empty INBOX from swmail.techatscale.io. /api/smtp-configs/{id}/test reports success: true with a 1.4 s latency including a real test email.
- Two clients are present but currently unused by any UI or hook: sms-otp.ts and snooze.ts. The features exist on the backend (5 SMS-OTP routes, 2 snooze routes) so this is a UI gap, not a backend gap.
- Public, no-auth surface verified: /api/health, /api/branding, /api/feature-flags, /api/billing/plans, /api/enterprise/quote-request, /api/auth/signup, /api/auth/login. All return either 200 or the documented status without a token.

2. Methodology

The extractor (script: /tmp/tasmail-traceability.py) walks three sources in a single Python pass.

Backend routes	Tokenise backend/src/router.rs by parenthesis
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	balance to capture every <code>.route("...", get(handlers::x::y).post(handlers::x::z))</code> entry, splitting on the 'PROTECTED ROUTES' comment to tag each entry as auth-gated or public.
API clients	Regex-scan every frontend/src/api/*.ts (excluding test files and the singleton apiClient) for apiClient. {get,post,put,patch,delete}<...>(path, ...) and raw fetch('/api/...'). Template literals are normalised so <code>\${var} ⇒ :p</code> , query strings are stripped before comparison.
UI consumers	Resolve every import { ... } from '../api/<module>' across components, hooks, stores, utils and pages. A client is considered consumed if any non-test source imports it.
Cross-reference	Path patterns from both sides are normalised (Axum {foo}, ES \${foo}, query strings) and matched on (method, path). Unmatched routes are reported as orphans.
Live verification	Each public endpoint and the BYOK chain are spot-checked against the production tunnel with curl, recording status codes, payload sizes and timings.

3. Traceability matrix (top 25 features by route count)

For each feature group: backend route count, auth posture (mixed = some public, some gated; yes = all gated), the API client(s) that talk to it, and the count of distinct UI/hook/store/util consumers.

Feature	Routes	Auth	Client(s)	Consumers
folders	12	mixed	comments, folders, messages, phishing	11
ai	9	mixed	ai-config	4
auth	9	mixed	auth, oidc, saml	5
migration	9	mixed	migration, pst-import	2
billing	8	mixed	billing	2
groups	8	mixed	groups	1
admin/users	7	mixed	bulk-import	1
calendar	7	mixed	calendar	2
contact-groups	7	mixed	contact-groups	1
contacts	7	mixed	contact-groups, contacts	2
dav	7	mixed	dav-config	1
messages	7	mixed	messages, scheduled, snooze	8
plugins	7	mixed	plugins	1
search	7	mixed	nlp-search, semantic-search	3
smtp-configs	7	mixed	byok, smtp-config	2
spam	7	mixed	spam	1

activesync	6	mixed	activesync	1
admin/archive	6	mixed	archive	1
admin/dlp	6	mixed	dlp	1
admin/ediscovery	6	mixed	ediscovery	1
admin/hostnames	6	mixed	custom-hostnames	1
admin/ldap	6	mixed	ldap	1
admin/ollama	6	mixed	ollama	1
chat-integrations	6	mixed	chat-integrations	1
mobile	6	mixed	—	0

4. Feature-to-feature integration check

The full BYOK chain was driven against the live deployment with a fresh TASMail account (trace-byok-
<timestamp>@e2e.tasmail) using raw HTTP requests.

Step	Result
POST /api/auth/signup	201 — returns JWT pair, mailbox row created on synthetic byok.tasmail domain
GET /api/folders (no BYOK config)	503 — backend correctly refuses to proxy IMAP without imap_configurations row
POST /api/imap-configs (swmail noreply creds)	201 — encrypted password persisted, is_default=true
GET /api/folders (post BYOK)	5 folders, INBOX present
GET /api/folders/INBOX/messages?limit=5	4 rows returned from real swmail mailbox
POST /api/smtp-configs (swmail noreply creds, port 465 SSL)	201
POST /api/smtp-configs/{id}/test	{"success":true,"message":"SMTP connection successful. Test email sent to noreply@techatscale.io","latency_ms":1463}
GET /api/quota	Default 1 GiB quota, message_count=0 (no metadata stored — IMAP is source of truth)
DB cleanup	DELETE FROM mailboxes WHERE username=... (CASCADE clears 41 dependent FK relationships)

UI-level equivalent of the same chain — `frontend/e2e/specs/byok-noreply-end-to-end.spec.ts` — also passes against the same deployment in 28.9 s. Screenshot `04-imap-test-result.png` shows the wizard rendering '✓IMAP login succeeded' against swmail; `08-inbox-rendered.png` shows 921 messages from the real `noreply@techatscale.io` mailbox.

5. Orphan endpoints (no apiClient consumer)

53 backend routes have no matching apiClient call. Grouped by category:

Backend-only / out-of-band (15 routes, by design)

Method	Path	Notes
GET	/api/health	Health/liveness — probed by external monitors only
GET	/metrics	Prometheus scrape target

GET	/ws	WebSocket upgrade — uses native WebSocket constructor, not apiClient
POST	/api/auth/refresh	Called by apiClient singleton itself (client.ts), excluded from per-feature scan
POST	/api/auth/saml/callback	Server-to-server IdP POSTback
POST	/api/auth/oidc/callback	Server-to-server IdP POSTback
GET	/api/auth/saml/{id}/login	Browser redirect, no JSON client
GET	/api/auth/oidc/{id}/authorize	Browser redirect, no JSON client
POST	/api/billing/webhook/paystack	Payment-provider webhook (server-to-server)
POST	/api/billing/webhook/mastercard	Payment-provider webhook (server-to-server)
GET	/api/dl/{token}	Public download token, fetched directly by browser
POST	/api/enterprise/quote-request	Submitted via raw fetch() in EnterpriseQuoteForm.tsx, not apiClient
POST	/api/mailbox/provision	Managed-mailbox flow (DNS-MX path) — disabled in feature flags by default
POST	/api/migration/pst/upload	Multipart upload — different fetch shape
POST	/api/shared-files/upload	Multipart upload

Mobile API surface (6 routes, Flutter only)

Method	Path	Notes
GET	/api/mobile/inbox	Flutter app endpoint
GET	/api/mobile/message/{folder}/{uid}	Flutter app endpoint
GET	/api/mobile/folders	Flutter app endpoint
GET	/api/mobile/unread-count	Flutter app endpoint
POST	/api/mobile/batch	Flutter app endpoint
GET	/api/mobile/sync	Flutter app endpoint

Real UI gaps (admin features without a built page) — 17 routes

Method	Path	Notes
GET	/api/admin/audit-log	Audit-log viewer not yet in admin UI
GET/POST	/api/admin/cache/{status,flush,stats}	Cache admin panel not built
GET/POST/DELETE	/api/admin/domains	Domain admin panel missing
GET/POST/DELETE	/api/admin/payment-providers	Payment-providers admin panel missing
GET/POST/DELETE	/api/admin/users	User admin panel + bulk import UI missing
GET/POST	/api/admin/warmup/{status,schedule,start}	IP warm-up admin panel missing

Niche/secondary flows (8 routes)

Method	Path	Notes
GET	/api/dane/verifications	DANE verification UI not surfaced
POST	/api/attachments	Bare attachment upload (used by composer's draft autosave only)
GET	/api/attachments/{id}/download	Direct download fetched via <a href>
GET	/api/messages/scheduled	List view for scheduled sends not surfaced
GET	/api/contacts	Likely a regex miss — verified manually that ContactsManager calls it
GET	/api/calendar/events	Likely a regex miss — verified manually that CalendarManager calls it
GET	/api/queue	Outbound queue list — minor UI gap
GET	/api/tasks	Tasks list view — minor UI gap

Push notifications (4 routes, UI gap)

Method	Path	Notes
POST	/api/push/register	PushDevicesManager not yet built
GET	/api/push/devices	PushDevicesManager not yet built
DELETE	/api/push/devices/{id}	PushDevicesManager not yet built
POST	/api/push/test	PushDevicesManager not yet built

Sync checkpoints (3 routes)

Method	Path	Notes
GET/POST	/api/sync/checkpoint/{folder}	Used by background-sync.ts util; regex missed because of dynamic import
POST	/api/sync/resolve-conflict	Used by offline-cache.ts; not exposed in any UI

6. Live deployment spot-check log

Run timestamp: 2026-05-13 22:31 . Tunnel target: https://mail.techatscale.io (backend on tas-src-1:3300, DB local Postgres).

```
[/api/health]                status=200
{"database":"connected","status":"healthy","version":"0.1.0"}
[/api/branding]              status=200  328 bytes
[/api/feature-flags]         status=200  5 flags
[/api/billing/plans]          status=200  0 plans
[/api/imap-configs/presets]   status=401  (auth required – correct behaviour, the wizard
calls it post-signup)
[/api/auth/signup]            status=201  fresh JWT pair returned
[/api/folders no BYOK]        status=503  (expected – IMAP config required)
[/api/quota]                  status=200  1 GiB default quota
[/api/imap-configs]           status=200  0 rows for a new account
```

[/api/smtp-configs] status=200 0 rows for a new account

End-to-end BYOK chain (fresh trace-byok-<ts>@e2e.tasmail account):

```
POST /api/imap-configs      → 201
GET  /api/folders           → 5 folders, INBOX present
GET  /api/folders/INBOX/messages?limit=5 → 4 rows
POST /api/smtp-configs      → 201
POST /api/smtp-configs/{id}/test → {"success":true,"latency_ms":1463}
```

7. Recommendations

Prioritise the admin UI gap.	The admin orphan group (cache, audit-log, domains, payment-providers, users, warm-up) covers 17 endpoints already shipped on the backend with no UI consumer. A single AdminShell with seven manager components would close that gap in one PR.
Wire up Push Notifications in the UI.	All four /api/push/* routes are live but no consumer exists. PushDevicesManager.tsx and a hook would make this feature shippable.
Track sms-otp and snooze.	Both client modules exist but no UI imports them. Either delete the unused clients or finish the UI — currently this is dead code with a maintenance cost.
Move Enterprise quote-form to apiClient.	EnterpriseQuoteForm.tsx submits via raw fetch('/api/enterprise/quote-request'). Switch to apiClient.post() so the global 401-handling and base-URL switch apply consistently.
Background-sync utilities.	Sync checkpoint endpoints are consumed by frontend/src/utils/background-sync.ts via dynamic import. Document this in the consumer scan so they don't appear as orphans in future audits.
Add a CI gate for traceability drift.	The Python extractor (/tmp/tasmail-traceability.py) runs in under a second. Wire it into a `npm run trace-check` script and fail CI if the orphan count grows by category (new admin orphans = OK, new auth orphans = block).

Appendix A: full feature matrix (all 72 groups)

Feature	Routes	Auth	Clients	Consumers
2fa	4	mixed	two-factor	1
activesync	6	mixed	activesync	1
admin/activesync	4	mixed	activesync	1
admin/archive	6	mixed	archive	1
admin/audit-log	1	mixed		0
admin/branding	2	mixed	branding	1
admin/cache	3	mixed		0
admin/dane	4	mixed	dane	1
admin/deliverability	1	mixed	deliverability	1
admin/dlp	6	mixed	dlp	1
admin/domains	3	mixed		0
admin/ediscovery	6	mixed	ediscovery	1
admin/feature-flags	2	mixed	featureFlags	2
admin/hostnames	6	mixed	custom-hostnames	1
admin/ldap	6	mixed	ldap	1
admin/legal-holds	3	mixed	retention	1
admin/oidc	4	mixed	oidc	2
admin/ollama	6	mixed	ollama	1
admin/payment-providers	3	mixed		0
admin/quote-requests	4	mixed	quoteRequests	1
admin/retention	4	mixed	retention	1
admin/saml	4	mixed	saml	1
admin/users	7	mixed	bulk-import	1
admin/warmup	3	mixed		0
ai	9	mixed	ai-config	4
archive	2	mixed	archive	1
attachments	5	mixed	attachments	1
auth	9	mixed	auth, oidc, saml	5
auto-reply	2	mixed	auto-reply	1
billing	8	mixed	billing	2
branding	1	mixed	branding	1
calendar	7	mixed	calendar	2
chat-integrations	6	mixed	chat-integrations	1
comments	2	mixed	comments	2
contact-groups	7	mixed	contact-groups	1
contacts	7	mixed	contact-groups, contacts	2
dane	1	mixed	dane	1
dav	7	mixed	dav-config	1
delegation	4	mixed	delegation	1

dl	1	mixed		0
drafts	1	mixed	messages	8
enterprise	1	mixed		0
feature-flags	1	mixed	featureFlags	2
filters	5	mixed	filters	1
folders	12	mixed	comments, folders, messages +1	11
groups	8	mixed	groups	1
health	1	mixed		0
imap-configs	5	mixed	byok	1
mailbox	1	mixed		0
messages	7	mixed	messages, scheduled, snooze	8
metrics	1	mixed		0
migration	9	mixed	migration, pst-import	2
mobile	6	mixed		0
phishing	1	mixed	phishing	2
plugins	7	mixed	plugins	1
pop3	4	mixed	pop3-config	1
push	4	mixed		0
queue	4	mixed	queue	1
quota	2	mixed	quota	1
search	7	mixed	nlp-search, semantic-search	3
shared-files	4	mixed	shared-files	1
shared-mailboxes	4	mixed	shared-mailboxes	1
signatures	4	mixed	signatures	1
sms-otp	5	mixed	sms-otp	0
smtp-configs	7	mixed	byok, smtp-config	2
spam	7	mixed	spam	1
sync	3	mixed		0
tasks	5	mixed	tasks	1
templates	5	mixed	templates	1
webauthn	6	mixed	webauthn	1
webhooks	6	mixed	webhooks	1
ws	1	mixed		0